Further Maths Revision Notes

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Chapter 1

Core Pure Year I

1.1 Complex Numbers I

1.1.1 The Imaginary Constant

Using only the real numbers, there is no number x that satisfies the equation $x^2 + 1 = 0$. To satisfy this polynomial, you must first define a square root of -1. This constant is called the imaginary constant i, defined such that $i^2 = -1$.

Using the imaginary constant, complex numbers can be defined as $a + bi$, where $a, b \in \mathbb{R}$. The set of all numbers in the form $a + bi$ is called the set of complex numbers, written $\mathbb{C}$.

The functions $\text{Re}(z)$ and $\text{Im}(z)$ (sometimes written as $\Re(z)$ and $\Im(z)$) take the real part and imaginary part of a complex number respectively. For example,

$$\text{Re}(a + bi) \equiv \Re(z) := a, \ \ a, b \in \mathbb{R} \tag{1.1}$$

$$\text{Im}(a + bi) \equiv \Im(z) := bi, \ \ a, b \in \mathbb{R} \tag{1.2}$$

1.1.2 Complex Arithmetic

Complex numbers can be added and multiplied similarly to how binomials would be. Given two complex numbers $w := a + bi$ and $z := c + di$, where $a, b, c, d \in \mathbb{R}$, the following rules define complex addition and multiplication:

$$w + z = (a + bi) + (c + di) = (a + c) + (b + d)i \tag{1.3}$$

$$w \times z = (a + bi)(c + di) = (ac - bd) + (ad + bc)i \tag{1.4}$$

1.1.3 Complex Conjugates

The complex conjugate of a complex number of z, written as z^* is z with a flipped sign before the imaginary part.

$$(a + bi)^* := a - bi, \ \ a, b \in \mathbb{R} \tag{1.5}$$

From this definition, we can see that the sum and product of a complex number and its conjugate are both real. Let $z := a + bi$, where $a, b \in \mathbb{R}$.

$$z + z^* = 2a \in \mathbb{R} \tag{1.6}$$

$$zz^* = a^2 + b^2 \in \mathbb{R} \tag{1.7}$$

Using the multiplication rule in equation 1.7, a fraction of complex numbers can be 'realised' by multiplying by the complex conjugate of the denominator over itself. Realising the denominator means rewriting a fraction of complex numbers as a fraction with a real denominator.

$$\begin{aligned} \frac{a+bi}{c+di} &= \frac{a+bi}{c+di} \times \frac{c-di}{c-di} \\ &= \frac{(a+bi)(c+di)}{c^2+d^2} \\ &= \frac{(ac-bd)+(ad+bc)i}{c^2+d^2} \end{aligned} \tag{1.8}$$

1.1.4 Complex Roots of Polynomials

All nth degree polynomials (polynomials with a power of x up to x^n) have exactly n roots, counting complex roots and algebraic multiplicities. Therefore, if an equation does not cross $y = 0$ on a Cartesian graph, it must have other, complex roots. The simplest example of this is the quadratic $x^2 + 1 = 0$.

$$x^2 = -1$$

$$x = \pm i$$

For quadratics, whenever the discriminant ($\Delta := b^2 - 4ac$ for a quadratic $ax^2 + bx + c$) is less than 0, the quadratic has complex roots, since the quadratic formula (B.1) can be rewritten as

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

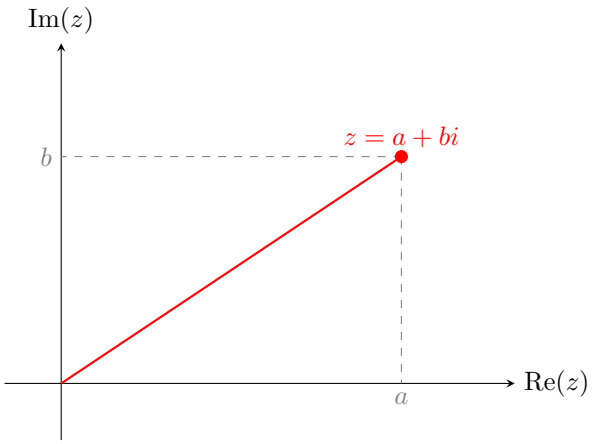
which would give a complex number whenever $\Delta < 0$, since $\sqrt{x}$ is complex iff $x < 0$ ($x \in \mathbb{R}$).

If a polynomial has a complex root α, it must also have the complex conjugate root α^*. For example, if a quadratic has a root $\alpha = 2 + 3i$, then the other root must be $\alpha^* = 2 - 3i$. For a cubic, there can only be one pair of roots, thus the third root cannot be a complex root, since it would have to have a fourth root to form a conjugate pair. Thus, the third root of a cubic must a real number.

1.2 Argand Diagrams

1.2.1 Complex Numbers on Argand Diagrams

An Argand diagram can be thought of similarly to a Cartesian graph; complex numbers in the form $x + iy$ can be plotted like (x, y) can on a Cartesian graph. Instead of an x axis and a y axis, there are instead a real axis and an imaginary axis, used to plot the real part and imaginary part of complex numbers. An Argand diagram looks like this:



Where the point $z = a + bi$ is drawn where the point (a, b) would be drawn on a Cartesian graph.

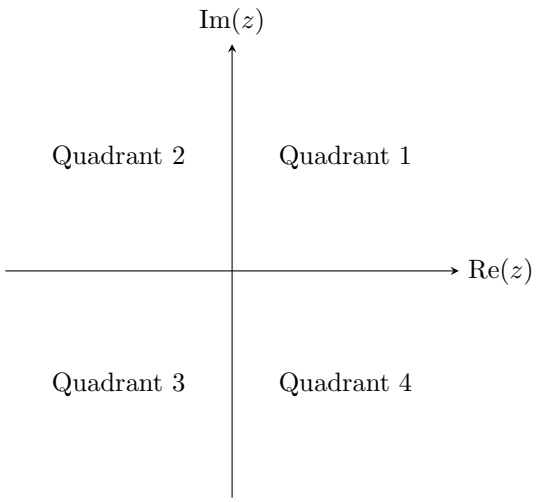
1.2.2 Modulus-Argument Form

The modulus-arguemnt form of a complex number is similar to the polar form of a point on a Cartesian graph. It defines the point using its distance from the origin, and the angle it makes with the positive half of the real axis, in radians. The modulus of z is written as $|z|$, which is calculated using Pythagoras' Theorem:

$$|z| = \sqrt{\text{Re}^2(z) + \text{Im}^2(z)} \tag{1.9}$$

where $\text{Re}^2(z)$ and $\text{Im}^2(z)$ mean $(\text{Re}(z))^2$ and $(\text{Im}(z))^2$ respectively. The angle that z makes with the positive real number line is called the argument, and is positive, going anti-clockwise if $\text{Im}(z) \geq 0$, and negative, going clockwise if $\text{Im}(z) < 0$. The argument of z is written as $\arg(z)$.

An Argand diagram is split into four quadrants, called quadrant 1, 2, 3 and 4:



For values of z in the first and fourth quadrants ($\text{Re}(z) > 0$):

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) \tag{1.10}$$

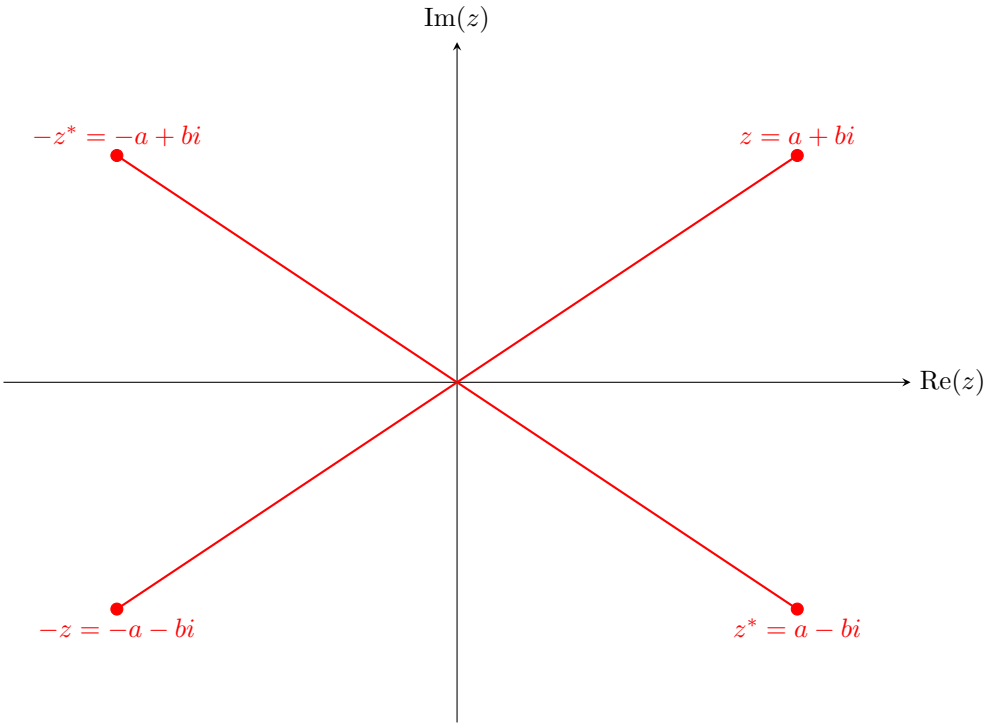
For values of z in the second quadrant:

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) + \pi \tag{1.11}$$

For values of z in the third quadrant:

$$\arg(z) = \arctan\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) - \pi \tag{1.12}$$

On an Argand diagram, the complex conjugate of a number is itself mirror in the line $\text{Im}(z) = 0$ (see subsection 1.2.3):



1.2.3 Loci and Regions in Argand Diagrams

1.3 Series I

1.4 Roots of Polynomials

1.4.1 Sums of Roots

The roots of polynomials can be related using Vieta's formulae (see). The roots of a general polynomial are written as α, β, γ, δ, etc., and the coefficients are written as a, b, c, d, e, etc., where a is the coefficient of the greatest power of the polynomial, b is the coefficient of the next greatest power, etc. The sum of all roots is then written as $\sum \alpha \equiv \alpha + \beta + \cdots$. Multiple roots can be multiplied together and then summed, for example the sum of all pairs of roots $(\alpha\beta + \alpha\gamma + \cdots)$ would be written as $\sum \alpha\beta$. This would be the sum of *all* possible pairs of roots.

1.4.2 Linear Transformations of Roots

1.5 Volumes of Revolution

1.6 Matrices

1.7 Linear Transformations

1.8 Proof by Induction

1.9 Vectors

Chapter 2

Core Pure Year II

2.1 Complex Numbers II

2.2 Series II

2.3 Methods in Calculus

2.4 Polar Coordinates

2.5 Hyperbolic Function

2.6 Methods in Differential Equations

2.7 Modelling with Differential Equations

Chapter 3

Decision Year I

3.1 Algorithms

3.2 Basic Algorithms

3.3 Bubble Sort

3.3.1 Description

Bubble sort is an algorithm that sorts an unsorted list of data. It passes through the list, checking each pair of items to see if they need to be swapped to be in order. For example, if it were sorting a list of numbers into numerical order and it found the pair 2, 1, it would swap them. This is an example of a single bubble sort pass, with each comparison marked:

```
[3   5]  8   2   5   6   3
 3  [5   8]  2   5   6   3
 3   5  [8   2]  5   6   3
 3   5   2  [8   5]  6   3
 3   5   2   5  [8   6]  3
 3   5   2   5   6  [8   3]
 3   5   2   5   6   3   8
```

Now, the 8 is in the right place, and the bubble sort will do another pass, ignoring the 8, since it must be in the right place after 1 pass, since it is the largest number. In general, after n passes, the last n items are sorted. Bubble sort gets its name from this property of the largets elements 'bubbling' to the top (end) of the list. If the algorithm does a pass without swapping any items, it must have already finished sorting the list, so it will return early.

Example of bubble sort:

Pass 1:	3	5	8	2	5	6	3
Pass 2:	3	5	2	5	6	3	**8**
Pass 3:	3	2	5	5	3	**6**	**8**
Pass 4:	2	3	5	3	**5**	**6**	**8**
Pass 6:	2	3	3	**5**	**5**	**6**	**8**
Pass 7:	2	3	**3**	**5**	**5**	**6**	**8**
No more swaps; sort complete.							

Notice that pass 6 is the same as pass 7, as only on pass 7 has the algorithm not performed any swaps.

3.3.2 Order

Theorem 3.3.1. *Bubble sort has an order of* $\mathcal{O}(n)$

Proof. Let n be the number of elements in the list to be sorted. This means that (in the worst-case scenario) the bubble sort will have to do $i - 1$ pass to guarantee that every item has been sorted (since after the ith pass, i items have been sorted, $n - i$ items will be left to sort, the ith item must also be sorted). In the ith pass, $i - 1$ items will have been sorted, so the total number

of comparisons will be:

$$\sum_{i=1}^{n}(n-i) = \frac{n(n-1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$$

Taking the fastest-growing term of $\frac{1}{2}n^2 + \frac{1}{2}n$ (the $\frac{1}{2}n^2$ term) and discarding the constant coefficient ($\frac{1}{2}$), this gives n^2, thus the order of the bubble sort algorithm is $\mathcal{O}(n^2)$. □

3.3.3 Implementation

In pseudocode, this is an example implementation of a bubble sort algorithm:

```
// Sorts a list using bubble sort; returns sorted list
// Example:
//     Input:  [3, 2, 7, 3, 5]
//     Output: [2, 3, 3, 5, 7]
//     Explanation: The unsorted input is sorted so the the numbers are returned in order
FUNC bubble_sort(list):            // Takes an unsorted list as input
    VAR n = list.UPPER_BOUND

    FOR (i = 0 TO n):                     // Goes through each pass
        VAR hasSwapped = FALSE
        FOR (j = 0 TO n - i):             // Goes through each value in the list
            IF (list[i] > list[j + 1]):   // Checks if items need swapping
                hasSwapped = TRUE
                VAR temp = list[j + 1]    // Swaps items
                list[j + 1] = list[i]
                list[i] = temp
            END IF
        END FOR

        IF (NOT hasSwapped):   // If it doesn't find any items to swap, the list must be sorted
            OUTPUT("No more swaps; sort complete.")
            RETURN list        // Return the sorted list early, since there are no more swaps
        END IF
    END FOR

    RETURN list                // Returns sorted list
END FUNC
```

3.4 Quick Sort

3.4.1 Description

A quick sort is a recursive sorting algorithm. It begins by choosing a 'pivot' in the middle of the list (the $\lceil \frac{n}{2} \rceil$th element). Then, it splits the list into two smaller sub-lists, the left and the right. The left sub-list will contain all elements less than or equal to the pivot, and the right list will contain all elements greater than the pivot. The pivot is then left in the middle and is now sorted. Then, each sub-list with a length greater than 1 is sorted using quick sort, hence quicksort is recursive.

This is an example of quick sort (the pivot will be in brackets, and then sorted elements will be bold):

Pass 1:	3	5	8	(2)	5	6	3
Pass 2:	**2**	3	5	8	(5)	6	3
Pass 3:	**2**	3	(3)	5	**5**	8	(6)
Pass 4:	**2**	3	(3)	5	**5**	**6**	8

All sub-lists contain only 1 item; sort complete.

3.4.2 Order

Theorem 3.4.1. *Quick sort has an order of* $\mathcal{O}(n^2)$.

Proof. The worst case scenario for quicksort is where the pivot consistently falls on the maximum/minimum pivot, leaving the sub-lists with lengths 0 and $n-1$. This means that the time taken to sort the list is the time taken for the nth pass plus the time taken to sort $n-1$ items. Lorem ipsum; QED □

3.4.3 Implementation

3.5 Bin-Packing Algorithms

3.6 Order of Algorithms

The order of an algorithm is how long it takes an algorithm to finish, based on the size of its input n, in the worst-case scenario. This is also called its time complexity, and is written as $\mathcal{O}(f(n))$, where $f(n)$ is a function of the algorithm's input size.

For example, the linear search algorithm looks through each element in a list one by one, checking to try to find a specific item. The worst case for linear search is if the item being looked for is either not in the list, or is the final item in the list, meaning that the algorithm has to check all n elements. This means that linear search has an order of $\mathcal{O}(n)$.

The order of an algorithm can be used to find the time taken for an algorithm to finish based on other recorded times. For example, if an $\mathcal{O}(n)$ algorithm takes 5ms to finish with 100 items, it will take 10ms to finish with 200 items.

Common orders are $\mathcal{O}(1)$ (constant time), $\mathcal{O}(n)$ (linear time), $\mathcal{O}(n^2)$ (quadratic time), and $\mathcal{O}(n^3)$ (cubic time). Quadratic, cubic, etc. time is also called polynomial time. An algorithm is considered more efficient if its order scales slower with a bigger input. For example, if an algorithm runs in constant time ($\mathcal{O}(1)$), then that is considered very good, since with an arbitrarily big input, it will still run in the same amount of time. On the other hand, an algorithm with an order of $\mathcal{O}(2^x)$ (exponential time) would be considered inefficient and slow, since it scales very quickly as the input grows, making it very slow for large inputs.

3.7 Graphs

3.8 Algorithms on Graphs

3.9 Route Inspection

3.10 The Travelling Salesman Problem

3.11 Linear Programming

3.12 The Simplex Algorithm

3.13 Critical Path Analysis

Chapter 4

Decision Year II

4.1 Transportation Problems

4.2 Allocation Problems

4.3 Flows in Networks I

4.4 Flows in Networks II

4.5 Dynamic Programming

4.6 Game Theory

4.7 Recurrence Relations

4.8 Decision Analysis

Chapter 5

Further Mechanics Year I

5.1 Momentum and Impulse

5.1.1 Momentum

Momentum is the property of an object given by its mass and velocity; the more massive an object is and the faster it is moving, the more momentum it has. A car moving slowly won't have much momentum since its velocity is low, but if it is moving quickly it will. A bullet thrown by hand wouldn't have much momentum, since it is very small and not very fast moving, but when shot, it would have a lot of momentum, since it is moving very fast, despite being small.

Since momentum is proportional to velocity and momentum, the equation is given as:

$$\mathbf{p} = m\mathbf{v} \tag{5.1}$$

Where $\mathbf{p}$ is momentum, m is the mass of the object, and $\mathbf{v}$ is the velocity. See equation B.43.

Units

In SI units, momentum is in kilograms (kg), and momentum is in metres per second (ms^{-1}). Since the two values are multiplied, this means that the units of momentum must be kilogram metres per second (kgms^{-1}).

However, since newtons are defined as kilogram metres per second squared (kgms^{-2}), newton seconds (Ns) would be kilogram metres per second (kgms^{-1}), which is the unit for momentum, so momentum can either be written in kgms^{-1} or in Ns.

Force/Time Definition

Since the units of momentum can be written as newton seconds (the unit of force $\times$ the unit of time), it looks like it might be possible to write momentum as a product of force and time.

Theorem 5.1.1. *Momentum is the product of force and time:*

$$\mathbf{p} = \mathbf{F}t \tag{5.2}$$

Proof. We start the proof with the following identities:

$$\mathbf{F} = m\mathbf{a} \tag{5.3}$$

$$\mathbf{a} = \frac{\mathbf{v}}{t} \tag{5.4}$$

$$\mathbf{p} = m\mathbf{v} \tag{5.5}$$

From this, $\mathbf{F}$ can be written in terms of m, $\mathbf{v}$, and t by substituting equation 5.4 into equation 5.3 we get:

$$\mathbf{F} = m\frac{\mathbf{v}}{t}$$

Multiplying both sides by t gives:

$$\mathbf{F}t = m\mathbf{v}$$

And finally, the $m\mathbf{v}$ can be replaced with the identity in equation 5.5 to get the result:

$$\mathbf{p} = \mathbf{F}t$$

□

This means that the momentum given to a particle after exerting a force $\mathbf{F}$ for t time will be $\mathbf{F}t$.

5.1.2 Impulse

Impulse ($\mathbf{I}$) is the change of momentum between two moments, i.e.:

$$\mathbf{I} = \Delta\mathbf{p} = m(\mathbf{v} - \mathbf{u}) \tag{5.6}$$

Where $\mathbf{v}$ is final momentum, and $\mathbf{u}$ is the initial momentum.

5.1.3 Momentum and Impulse as Vectors

5.2 Work, Energy and Power

5.3 Elastic Strings and Springs

5.4 Elastic Collisions in One Dimension

5.5 Elastic Collisions in Two Dimensions

Chapter 6

Further Mechanics Year II

6.1 Circular Motion

6.2 Centre of Mass of Plane Figures

6.3 Further Centres of Mass

6.4 Kinematics

6.5 Dynamics

Appendix A

Appendix: Formula Sheet

This formula sheet contains the formulae from the Pearson Edexcel formula sheet. See original here.

A.1 Pure Mathematics

A.1.1 Summations

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1) \tag{A.1}$$

$$\sum_{r=1}^{n} r^3 = \frac{1}{6}n^2(n+1)^2 \tag{A.2}$$

A.1.2 Matrix Transformations

Anticlockwise rotation through θ about O:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{A.3}$$

Reflection in the line $y = (\tan\theta)x$:

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \tag{A.4}$$

A.1.3 Area of a Sector

$$A = \frac{1}{2}\int r^2 d\theta \quad \text{(polar coordinates)} \tag{A.5}$$

A.1.4 Complex Numbers

$$[r\cos\theta + i\sin\theta]^n = r^n(\cos n\theta + i\sin n\theta) \tag{A.6}$$

The roots of $z^n = 1$ are given by

$$z = e^{\frac{2\pi k i}{n}}, \text{ for } k = 0, 1, 2, \ldots, n-1 \tag{A.7}$$

A.1.5 Maclaurin's and Taylor's Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^r}{r!}f^{(r)}(0) + \cdots \tag{A.8}$$

$$e^x = \exp(x) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^r}{r!} + \cdots \quad \text{for all } x \tag{A.9}$$

$$\ln(n+1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^r\frac{x^r}{r} + \cdots \quad (-1 < x \leq 1) \tag{A.10}$$

$$\cos x = 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^{r+1}\frac{x^{2r+1}}{(2r+1)!} + \cdots \quad \text{for all } x \tag{A.11}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^r\frac{x^{2r}}{(2r)!} \quad \text{for all } x \tag{A.12}$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^r\frac{x^{2r+1}}{2r+1} + \cdots \quad (-1 \leq x \leq 1) \tag{A.13}$$

A.1.6 Vectors

The resolved part of $\mathbf{a}$ in the direction of $\mathbf{b}$ is

$$\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}} \tag{A.14}$$

Vector Product

$$\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\hat{\mathbf{n}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{bmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{bmatrix} \tag{A.15}$$

$$\mathbf{a}\cdot(\mathbf{b}b\times\mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \mathbf{b}(\mathbf{c}\times\mathbf{a}) = \mathbf{c} = (\mathbf{a}\times\mathbf{b}) \tag{A.16}$$

If A is the point with position vector $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and the direction vector $\mathbf{b}$ is given by $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, then the straight line through A with direction vector $\mathbf{b}$ has Cartesian equation:

$$\frac{x-a_1}{b_1} = \frac{y-a_2}{b_2} = \frac{z-a_3}{b_3} = \lambda \tag{A.17}$$

The plane through A with normal vector $\mathbf{n} = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$ has Cartesian equation:

$$n_1x + n_2y + n_3z + d = 0 \text{ where } d = -\mathbf{a}\cdot\mathbf{n} \tag{A.18}$$

The plane through non-collinear points A, B and C has vector equation:

$$\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b}-\mathbf{a}) + \mu(\mathbf{c}-\mathbf{a}) = (1-\lambda-\mu)\mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c} \tag{A.19}$$

The plane through A and parallel to $\mathbf{b}$ and $\mathbf{c}$ as vector equation:

$$\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c} \tag{A.20}$$

The perpendicular distance of (α, β, γ) from $n_1x + n_2y + n_3z + d = 0$ is:

$$\frac{|n_1\alpha + n_2\beta + n_3\gamma + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} \tag{A.21}$$

A.1.7 Hyperbolic Functions

$$\cosh^2 x - \sinh^2 x = 1 \tag{A.22}$$

$$\sinh 2x = 2\sinh x \cosh x \tag{A.23}$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x \tag{A.24}$$

$$\cosh^{-1} = \ln\left\{x + \sqrt{x^2-1}\right\} \quad (x \geq 1) \tag{A.25}$$

$$\sinh^{-1} = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1) \tag{A.26}$$

A.1.8 Conics

Ellipses

Standard Form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{A.27}$$

Parametric Form:

$$(a\cos\theta, b\sin\theta) \tag{A.28}$$

Eccentricity

$$\begin{gathered} e < 1 \\ b^2 = a^2(1-e^2) \end{gathered} \tag{A.29}$$

Foci:

$$(\pm ae, 0) \tag{A.30}$$

Directrices:

$$x = \pm\frac{a}{e} \tag{A.31}$$

Asymptotes: none

Parabolas

Standard Form:

$$y = 4ax \tag{A.32}$$

Parametric Form:

$$(at^2, 2at) \tag{A.33}$$

Eccentricity:

$$e = 1 \tag{A.34}$$

Foci:

$$(a, 0) \tag{A.35}$$

Directrices:

$$x = -a \tag{A.36}$$

Asymptotes: none

Hyperbolae

Standard Form:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{A.37}$$

Parametric Forms:

$$(a\sec\theta, b\tan\theta) \tag{A.38}$$

$$(\pm a\cosh\theta, b\sinh\theta) \tag{A.39}$$

Eccentricity:

$$\begin{gathered} e > 1 \\ b^2 = a^2(e^2 - 1) \end{gathered} \tag{A.40}$$

Foci:

$$(\pm ae, 0) \tag{A.41}$$

Directrices:

$$x = \pm\frac{a}{e} \tag{A.42}$$

Asymptotes:

$$\frac{x}{a} = \pm\frac{y}{b} \tag{A.43}$$

Rectangular Hyperbolae

Standard Form:

$$xy = c^2 \tag{A.44}$$

Parametric Form:

$$\left(ct, \frac{c}{t}\right) \tag{A.45}$$

Eccentricity:

$$e = \sqrt{2} \tag{A.46}$$

Foci:

$$(\pm\sqrt{2}c, \pm\sqrt{2}c) \tag{A.47}$$

Directrices:

$$x + y = \pm\sqrt{2}c \tag{A.48}$$

Asymptotes:

$$x = 0,\ y = 0 \tag{A.49}$$

A.1.9 Differentiation

$f(x)$	$f'(x)$	
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	(A.50)
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	(A.51)
$\arctan x$	$\frac{1}{\sqrt{1+x^2}}$	(A.52)
$\sinh x$	$\cosh x$	(A.53)
$\cosh x$	$\sinh x$	(A.54)
$\tanh x$	$\operatorname{sech}^2 x$	(A.55)
$\operatorname{arsinh} x$	$\frac{1}{1+x^2}$	(A.56)
$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x^2-1}}$	(A.57)
$\operatorname{artanh} x$	$\frac{1}{1-x^2}$	(A.58)
		(A.59)

A.1.10 Integration (+ constant; $a > 0$here relevant)

$f(x)$	$\int f(x)dx$	
$\sinh x$	$\cosh x$	(A.60)
$\cosh x$	$\sinh x$	(A.61)
$\tanh x$	$\ln \cosh x$	(A.62)
$\frac{1}{\sqrt{a^2 - x^2}}$	$\sin^- 1\left(\frac{x}{a}\right) + c \quad (\lvert x\rvert < a)$	(A.63)
$\frac{1}{a^2 + x^2}$	$\frac{1}{a}\tan^- 1\left(\frac{x}{a}\right) + c$	(A.64)
$\frac{1}{x^2 - a^2}$	$\cosh^- 1(\frac{x}{a})$ or $\ln\left\{x + \sqrt{x^2 - a^2} + c \quad (x > a)\right\}$	(A.65)
$\frac{1}{\sqrt{a^2 + x^2}}$	$\sinh^{-1}\left(\frac{x}{a}\right)$ or $\ln\left\{x + \sqrt{x^2 + a^2}\right\} + c$	(A.66)
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + c \quad (\lvert x\rvert < a)$	(A.67)
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a-x}{a+x}\right\rvert + c$	(A.68)

A.1.11 Arc Length

$$s = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \quad \text{(Cartesian coordinates)} \tag{A.69}$$

$$s = \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \quad \text{(parametric form)} \tag{A.70}$$

$$s = \int \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad \text{(polar form)} \tag{A.71}$$

A.1.12 Surface Area of Revolution

$$s_x = 2\pi \int y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \quad \text{(Cartesian coordinates)} \tag{A.72}$$

$$s_x = 2\pi \int y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \quad \text{(parametric form)} \tag{A.73}$$

$$s_x = 2\pi \int r\sin\theta\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad \text{(polar form)} \tag{A.74}$$

A.2 Mechanics

A.2.1 Constant Acceleration

$$s = ut + \frac{1}{2}at^2 \qquad \mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 \tag{A.75}$$

$$s = vt - \frac{1}{2}at^2 \qquad \mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2 \tag{A.76}$$

$$v = u + at \qquad \mathbf{v} = \mathbf{u} + \mathbf{a}t \tag{A.77}$$

$$s = \frac{1}{2}(u + v)t \qquad \mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t \tag{A.78}$$

$$v^2 = u^2 + 2as \tag{A.79}$$

A.2.2 Centres of Mass

For uniform bodies:

$$\text{Triangular lamina: } \frac{2}{3} \text{ along median from vertex} \tag{A.80}$$

$$\text{Solid hemisphere, radius } r\text{: } \frac{3}{8}r \text{ from centre} \tag{A.81}$$

$$\text{Hemispherical shell: } \frac{2}{3} \text{ along median from vertex} \tag{A.82}$$

$$\text{Circular arc, radius } r \text{ angle at centre } 2\alpha\text{: } \frac{r\sin\alpha}{\alpha} \text{ from centre} \tag{A.83}$$

$$\text{Sector of circle, radius } r \text{ angle at centre } 2\alpha\text{: } \frac{2r\sin\alpha}{3\alpha} \text{ from centre} \tag{A.84}$$

Solid cone or pyramid of height h: $\frac{1}{4}h$ above the base on the line from the centre of the base to the vertex (A.85)

Conical shell of height h: $\frac{1}{3}h$ above the base on the line from the centre of the base to the vertex (A.86)

A.3 Probability and Statistics

A.3.1 Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{A.87}$$

$$P(A \cap B) = P(A) \times P(B|A) \tag{A.88}$$

A.3.2 Standard Deviation

$$\sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \tag{A.89}$$

A.3.3 Discrete Distributions

Binomial $\mathrm{B}(n,p)$

$P(X=x)$:

$$\binom{n}{x} p^x (1-p)^{n-x} \tag{A.90}$$

Mean:

$$np \tag{A.91}$$

Variance:

$$np(1-p) \tag{A.92}$$

Poisson $\mathrm{Po}(\lambda)$

$P(X=x)$:

$$e^{-\lambda}\frac{\lambda^x}{x!} \tag{A.93}$$

Mean:

$$\lambda \tag{A.94}$$

Variance:

$$\lambda \tag{A.95}$$

Sampling Distributions

For a random sample $X_1, X_2, \ldots, X_n$ of n independent observations from a distribution having mean μ and a variance σ^2:

$$\bar{X} \text{ is an unbiased estimator of } \mu, \text{ with } \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} \tag{A.96}$$

$$S^2 \text{ is an unbiased estimator of } \sigma^2, \text{ where } S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1} \tag{A.97}$$

For a random sample of n observations from $\mathrm{N}(\mu, \sigma^2)$:

$$\frac{\bar{X} - \mu}{\sigma} \sim \mathrm{N}(0, 1) \tag{A.98}$$

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1} \tag{A.99}$$

A.3.4 Distribution-Free (Non-Parametric) Tests

Contingency tables:

$$\sum \frac{(O_i - E_i)^2}{E_i} \text{ is approximately distributed as } \chi^2 \tag{A.100}$$

Appendix B

Non-Formula Sheet

B.1 Pure Mathematics

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{B.1}$$

B.1.1 Vieta's Formulae

For a quadratic $ax^2 + bx + c$ with roots α, β:

$$\sum \alpha = \alpha + \beta = -\frac{b}{a} \tag{B.2}$$

$$\alpha\beta = \frac{c}{a} \tag{B.3}$$

For a cubic $ax^3 + bx^2 + cx + d$ with roots α, β, γ:

$$\sum \alpha = \alpha + \beta + \gamma = -\frac{b}{a} \tag{B.4}$$

$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \tag{B.5}$$

$$\alpha\beta\gamma = -\frac{d}{a} \tag{B.6}$$

For a quartic $ax^4 + bx^3 + cx^2 + dx + e$ with roots α, β, γ, δ:

$$\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{b}{a} \tag{B.7}$$

$$\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a} \tag{B.8}$$

$$\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a} \tag{B.9}$$

$$\alpha\beta\gamma\delta = \frac{e}{a} \tag{B.10}$$

B.1.2 Integration

$$\int u\frac{du}{dx}dx = uv - \int v\frac{du}{dx}dx \tag{B.11}$$

$$\int \frac{f'(x)}{f(x)} = \ln|f(x)| + c \tag{B.12}$$

$$\tan x \qquad \ln|\sec x| + c \tag{B.13}$$

$$\cot x \qquad \ln|\sin x| + c \tag{B.14}$$

$$\operatorname{cosec} x \qquad -\ln|\operatorname{cosec} x + \cot x| = \ln\left|\tan\frac{1}{2}x\right| + c \tag{B.15}$$

$$\sec x \qquad \ln|\sec x + \tan x| = \ln\left|\tan\left(\frac{1}{2}x + \frac{1}{4}\pi\right)\right| + c \tag{B.16}$$

$$\sec^2 x \qquad \tan x + c \tag{B.17}$$

$$\tanh x \qquad \ln\cosh x + c \tag{B.18}$$

B.1.3 Binomial Series

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + b^n \quad (n \in \mathbb{N}) \tag{B.19}$$

Arithmetic Series

$$S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n[2a + (n-1)d] \tag{B.20}$$

Geometric Series

$$S_n = \frac{a(1-r^n)}{1-r} \tag{B.21}$$

$$S_\infty = \frac{a}{1-r} \tag{B.22}$$

B.1.4 Trigonometry: Small Angles

$$\sin\theta \approx \theta \tag{B.23}$$

$$\cos\theta \approx 1 - \frac{\theta^2}{2} \tag{B.24}$$

$$\tan\theta \approx \theta \tag{B.25}$$

B.1.5 Trigonometric Identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \tag{B.26}$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B \tag{B.27}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} \tag{B.28}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2} \tag{B.29}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} \tag{B.30}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2} \tag{B.31}$$

B.1.6 Differentiation from First Principles

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{B.32}$$

B.1.7 Numerical Solutions of Equations

The Newton-Raphson iteration for solving $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{B.33}$$

B.1.8 Numerical Integration

The Trapezium Rule:

$$\int_a^b y dx \approx \frac{1}{2} \left\{ h(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1}) \right\}, \text{ where } h = \frac{b-a}{n} \tag{B.34}$$

B.1.9 Further Numerical Integration

The mid-ordinate rule:

$$\int_a^b y dx \approx h \left(y_{\frac{1}{2}} + y_{\frac{3}{2}} + \cdots + y_{n-\frac{3}{2}} + y_{n-\frac{1}{2}} \right) \tag{B.35}$$

where $h = \frac{b-a}{n}$
Simpson's Rule:

$$\int_a^b y dx \approx \frac{1}{3} h \left\{ (y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1} + 2(y_2 + y_4 + \cdots + y_{n-2})) \right\} \tag{B.36}$$

B.1.10 Numerical Solutions to Differential Equations

For $\frac{dy}{dx} = f(x)$ and small h, recurrence relations are:
Euler's Method:

$$y_{n+1} = y_n + hf(x), \quad x_{n+1} = x_n + h \tag{B.37}$$

For $\frac{dy}{dx} = f(x, y)$:
Euler's Method:

$$y_{r+1} = y_r + hf(x_r, y_r), \quad x_{r+1} = x^r + h \tag{B.38}$$

Improved Euler's Method:

$$y_{r+1} = y_{r-1} + 2hf(x_r, y_r), \quad x_{r+1} = x_r + h \tag{B.39}$$

B.1.11 3D Matrices

The matrices for rotations (in three dimensions) though an angle θ about one of the axes are:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad \text{for the } x\text{-axis} \tag{B.40}$$

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \quad \text{for the } y\text{-axis} \tag{B.41}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{for the } z\text{-axis} \tag{B.42}$$

B.2 Mechanics

B.2.1 Momentum and Impulse

$$\mathbf{p} = m\mathbf{v} \tag{B.43}$$

$$\mathbf{p} = \mathbf{F}t \tag{B.44}$$

$$\mathbf{I} = m\Delta\mathbf{v} = m(\mathbf{v} - \mathbf{u}) \tag{B.45}$$

$$\mathbf{I} = \Delta\mathbf{p} = m\mathbf{v} - m\mathbf{u} \tag{B.46}$$

B.3 Probability and Statistics

Glossary

ampere The SI unit of electric current. 71

Argand diagram A 2D graph of the complex plane. 3, 4

argument The angle that a complex number makes with the positive real number line. It is written as $\arg(z)$. The argument is in the real interval $(-\pi, \pi]$. 3

candela The SI unit of luminous intensity. It is approximately the luminous intensity of a wax candle. 71

Cartesian graph A graph in the x-y plane. 2, 3, 70

complex conjugate A complex number with a flipped sign before its imaginary part. The complex conjugate of a complex number z is written z^*. 1, 2, 4, 69

complex number A number in the form $a + bi$, where $a, b \in \mathbb{R}$. The set of complex numbers is written $\mathbb{C}$. 1, 2, 3, 69, 70, 71

complex root A root x of an equation where $x \in \mathbb{C}$. It is often also used to mean that $x \notin \mathbb{R}$. 2

conjugate pair A pair of complex numbers, where one is the complex conjugate of the other. 2

cubic A 3rd degree polynomial, i.e. an equation in the form $x^3+bx^2+cx+d = 0$, where $a \neq 0$. 2

force A push or pull on an object. It has the SI units newtons symbol (N). See equations **??**. 44

imaginary constant The square root of -1, written i. 1

imaginary part The coefficient of the imaginary term of a complex number, i.e. where $z = a+bi,\ a, b \in \mathbb{R}$, the imaginary part of z is b. The imaginary part of a complex number z is written as $imag(z)$ or $\Im(z)$. 1, 3

kelvin The SI unit of temperature. A change of temperature of 1K is the same as a change of 1°C, but 0K is approximately -273°C. 71

kilogram The SI unit of mass. It appears to have an SI prefix (kilo), however due to historical reasons, it is considered a base unit. The gram still exists, defined as a thousandth of a kilogram. 43, 70, 71

metre The SI unit of length. 43, 70, 71

modulus The magnitude of a number, where $|x|$ is the modulus of x. For positive real numbers, the modulus is itself, for negative real numbers, the modulus is the positive version of itself ($|x| = -x,\ x \in \mathbb{R}^-$). For a complex number z, $|z| = \sqrt{\mathrm{Re}^2(z) + \mathrm{Im}^2(z)}$. 3

modulus-arguemnt form The form $r(\cos\theta + i\sin\theta) \equiv \mathrm{cis}\,\theta$. 3

mole The SI unit of quantity. 1mol is approximately the number of atoms of carbon-12 in 12g of carbon-12. It is defined as N_A^-1. 71

momentum . 43, 70

momentum The product of an object's momentum and momentum. It the SI units kilogram metres per second (kgms^{-1}), or newton seconds (Ns). See equations B.43, B.44. 43, 44

momentum Velocity is a vector quantity measuring the rate of change of position of an object. See equations **??**. 43, 44, 70

newton The SI unit of force. The newton is not a base unit, but instead is equivalent to a kilogram metre per second squared (kgms^{-2}). 43, 69, 70

polynomial An expression in the form $a_0 + a_1x^1 + a_2x^2 + \cdots + a_nx^n$. 1, 2, 6, 69, 70

quadrant A quarter of a graph; the 1st quadrant is the top right, the second is the top left, the third is the bottom left, and the fourth is the bottom right.. 3, 4

quadratic A second degree polynomial, i.e. an equation in the form $ax^2 + bx + c = 0$, where $a \neq 0$. When plotted on a Cartesian graph, it forms a parabola. Its roots are given by the quadratic formula, $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$. 2, 65

radian An angle unit, equal to the angle required to make the arc length of a unit circle equal 1. A whole circle is 2π radians; half a circle is π radians. The symbol for radians is either ^{c} or rad, as in π^{c} or π rad, or is often omitted. 3

real number A number on the real number line; a number in the interval $(-\infty, \infty)$. The set of real numbers is written $\mathbb{R}$. 1, 2, 70

real part The real term of a complex number, i.e. where $z = a + bi,\ a, b \in =$, the real part of z is a The real part of a complex number z is written as $\mathrm{Re}(z)$ or $\Re(z)$. 1, 3

realising the denominator The process of multiplying a complex fraction in the form $\frac{w}{z}$ by $\frac{z^*}{z^*}$ (where $w, z \in \mathbb{C}$) to make the denominator a real value without changing the value of the fraction.. 2

root A value that satisfies an equation, for example 1 and -1 are the roots of the equation $x^2 - 1 = 0$, since for $x = 1$ and $x = -1$, the equation holds. 2, 6, 56, 65, 70

second The SI unit of time. 43, 70, 71

SI unit The International System of units; the standard unit system used globally for maths and other STEM subjects. All SI units are made from the 7 base units: the metre (m), kilogram (kg), the second (s), the ampere (A), the kelvin (K), the mole (mol) and the candela (cd). All of the base units are derived from 7 universal constants: the speed of light (c), the hyperfine transition frequency of caesium-133, the Planck constant (h), the Avrogadro constant (N_A), the luminous efficacy of 540THz radiation, the Boltzmann constant (k), and the elementary change (e). 43, 69, 70, 71

time The progression of existence, represented as the fourth dimension of (3+1)D spacetime. It has the SI units seconds (s). 44